

Team and project selection form, COMPSCI 4ZP6

Please fill out, and submit by Sep 19, on Avenue, in PDF only.

Team members (minimum is 5, maximum is 7):

Full name (as seen on Avenue)	McMaster Email address	Student Number
Dason Wang	wangx647@mcmaster.ca	400476534
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Project information:

Project title	Battery Algorithm Standardized Testing Tool – Public Web Interface and Parallelized Cloud Implement
Supervisor name	Dr. Phillip Kollmeyer
Supervisor email	kollmeyp@mcmaster.ca
Link to GitHub repo (make sure the instructor and TAs are added as collaborators)	https://github.com/Dason-IsopodOverseer/Battery-Algorithm-Standardized-Testing-Tool

Project description (~100 to ~200 words, please stay within this page, use font size 10):

Battery state of charge (SOC) estimation requires specialized algorithms and standardized testing to determine which SOC estimation approach (possibly hundreds) is optimal. A centralized platform to crowd-source SOC algorithms and efficiently evaluate them will accelerate research to better develop battery products, benefiting researchers, engineers, and hobbyists alike.

We propose to create a public-facing website/app to make Dr. Kollmeyer's SOC algorithm comparison tool available to the general public. We'll select and configure a reliable hosting solution, ensuring scalability; port over the existing web interface from MATLAB to Python to mitigate licensing conflicts; add functionality to include user accounts, leader-boards, activity, performance visualization, etc.; and ultimately implement parallel cloud testing of user submitted algorithms, distributed across a server farm, for example, to reduce user wait times. Additionally, we seek to create documentation (manuals, tutorials, architectural design records) to ensure the project can be managed well into the future.

We expect to host the platform on scalable cloud solutions (AWS, Azure, or McMaster resources). The frontend will possibly use React with TypeScript, with cloud databases like Azure SQL or DynamoDB for user accounts and submissions. GitHub will manage version control, while Monday will support project management.